

The Hilbert-Transform Surface Channel in CS-DNO

Rationale, Evidence Gap, and Deletion Ablation

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1 Question and present status

The C27 CS-DNO represents the surface with the fixed feature stack

$$E_0(\eta) = \left[\eta, \eta^2, \eta^3, \partial_x \eta, \partial_x^2 \eta, |D|^{1/2} \eta, \mathcal{H}\eta \right]. \quad (1)$$

The final channel was inherited from the early broad CS-DNO feature bank. It was never isolated in an ablation. Its inclusion is therefore a plausible architectural hypothesis, not an experimentally established requirement of C27.

2 What transform is used?

For a real periodic surface, the spatial Hilbert transform is the Fourier multiplier

$$\widehat{\mathcal{H}\eta}(k) = -i \operatorname{sgn}(k) \widehat{\eta}(k). \quad (2)$$

It preserves the magnitude of every nonzero Fourier mode and shifts its phase by ninety degrees. It is an order-zero, nonlocal operator: unlike $\partial_x \eta$ or $\partial_x^2 \eta$, it does not amplify high wavenumbers. It also satisfies

$$\mathcal{H}\partial_x = |D|, \quad (3)$$

which connects it to the nonlocal multipliers that occur in water-wave operators.

This channel is unrelated to the strip-Hilbert transform used in constructing Tanaka initial conditions. Here it is only a deterministic input feature of the learned surface trunk.

3 Why was it plausible?

After the fixed feature construction, the shared surface trunk is pointwise in x . Pointwise powers of η cannot reconstruct the quadrature component $\mathcal{H}\eta$. Supplying it explicitly gives the trunk access to nonlocal phase information without asking a learned local map to synthesize a Fourier operator. A concise explanation, if the channel survives ablation, is:

We include the periodic Hilbert transform of the surface to expose nonlocal quadrature information to an otherwise pointwise surface-feature trunk. The channel is order zero, so it supplies phase information without the high-wavenumber amplification of derivative channels.

This explains what information the channel makes available. It does not show that the information improves the learned DNO.

4 What evidence do we actually have?

The historical minimal-feature experiment compared the complete feature bank against a model that simultaneously reduced the polynomial degree and disabled all spectral surface channels. The deletion regressed validation loss by a factor of approximately 2.3, establishing that the *feature bank as a whole* matters. It does not identify the contribution of $\mathcal{H}\eta$.

In particular:

- no run changed only the Hilbert channel;
- neither the analytic $G_0 + G_1$ backbone nor the three C27 loss terms requires an explicit $\mathcal{H}\eta$ input;
- the current evidence cannot distinguish a useful phase feature from a redundant inherited channel.

The defensible pre-ablation statement is therefore that $\mathcal{H}\eta$ is an unisolated, physics-motivated feature. It should not be described as load-bearing on the basis of the existing grouped ablation.

5 Hilbert-only deletion protocol

Run a one-variable comparison after fixing the full-row C27 optimizer-step budget. Both arms must use the same fresh seed, authenticated corpus, row order, normalization, batch size, optimizer schedule, and three loss terms.

Arm	Hilbert channel	All other settings
Control	enabled	locked C27 recipe
Deletion	disabled	identical to control

The trainer already exposes the required switch:

`--cs_use_hilbert` versus `--no-cs_use_hilbert`.

At the locked C27 dimensions, the control has 1,342,400 parameters and the deletion has 1,342,080. The difference is only 320 parameters, approximately 0.024%, so the comparison is effectively capacity matched.

Evaluate both held-out one-step error and the frozen case-by-case rollout panel. The rollout comparison must include nonfinite counts, raw and translation-aligned tail errors, and per-family worst cases. A validation-only win is insufficient if rollout stability regresses.

6 Decision rule

Retain $\mathcal{H}\eta$ only if the control shows a reproducible accuracy or stability benefit larger than run-to-run variation. If the deletion is equivalent or better, remove the channel and prefer the smaller, easier-to-defend feature stack. Until that comparison is complete, preserve C27 when reproducing the locked model, but mark the channel as unresolved rather than justified.